

FAKULTI SAINS KOMPUTER & TEKNOLOGI MAKLUMAT

BIC20904 – OBJECT-ORIENTED PROGRAMMING

FINAL REPORT

CENTRALIZED MULTI-BRANCH CLINIC MANAGEMENT SYSTEM

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CHAPTER 1

INTRODUCTION

1.1 Introduction of the Project

Oral health is a vital yet often overlooked aspect of overall well-being. Many people often neglect dental care. They only seek help when they feel pain in their teeth or gum. However, dental health should not be taken lightly. We must do regular checkups even when we feel our teeth appear to be healthy because hidden issues such as decay or gum disease can be detected early. Patients can avoid costly emergency procedures in the future by prioritizing prevention care such as routine scaling and examinations.

To support this mission of proactive oral health care, the Putra Dental Clinic Management System has been developed. Putra Dental Clinic is a professional dental service clinic that is dedicated to offering high quality oral care to the community. The clinic specializes in a wide range of dental treatment from general dentistry and preventive care to more complex procedures such as braces. Currently, Putra Dental Clinic operates through three main branches which are located in Parit Raja, Pagoh, and Kluang, Johor.

The Putra Dental Clinic Management System is designed to streamline the clinic's daily operations ensuring the patient records, appointment scheduling, treatment histories, and payment process are managed efficiently across its multiple locations. By implementing an organized Object-Oriented system, the clinic can provide better service for the patients and help them to stay consistent with their dental health care.

1.2 Problem Background

The current operational framework of Putra Dental Clinic which is divided into three branches located in Parit Raja, Pagoh and Kluang relies heavily on manual processes and disorganized patient record storage. This setup led to some difficulties in centralizing the management of medical information such as medical supplies and standardized pricing across its three branches. Since each branch operates independently with a manual system, data is not synchronized and can cause inconsistencies in patient billing and inventory management. Without a unified digital database, the management of the clinic faces a hard situation to track stock levels or ensure treatment or medicine price equally. Later, this can cause operational delays and financial problems.

Furthermore, the internal communication between dentist, receptionists and HR staff struggle to communicate manually regarding the appointment dates, dental treatment types and specific medical details. These communication gaps frequently cause dentists to have misleading or inaccurate information from the receptionist. Then the clinic will miss scheduled appointment dates or unprepared for specific procedures. In the end, the lack of a centralized digital system disrupt a smooth patient experience for those who visit multiple branches since their dental record cannot be accessed beyond their original branch. Lastly, inefficient and poorly structured workflows will negatively impact the quality of dental care services at Putra Dental Clinic.

1.3 Objectives of the Project

The main objective is to develop an integrated clinic management system that centralizes the operations for Putra Dental Clinic across its branches in Parit Raja, Pagoh, and Kluang. The system aims to enhance communication efficiency by providing a unified platform where dentists, receptionists, and HR can access real-time digital records to ensure that appointment schedules and treatment details are accurate and synchronized. Additionally, this project aims to eliminate the risk related with manual record keeping such as misleading or lost appointment information files and overlooked medical histories.

1.4 Scope of the Project

The scope of the Putra Dental Clinic Management System includes the development of a comprehensive platform designed to synchronize operations across the Parit Raja, Pagoh, and Kluang branches. The system enables patients to conveniently schedule, modify, and cancel dental appointments online, while receptionists can manage these visits through a real-time scheduling calendar. To ensure secure data access, the system supports a Login Module for various user roles, including Admins, Dentists, Receptionists, HR, and Patients using unique usernames and passwords. Furthermore, the system enables clinical staff to record and manage detailed patient information, including contact details, visit history, and treatment records to support continuous patient care..

The project also covers administrative and financial management features to improve clinic efficiency. Admin are granted full authority to create and manage an updated list of dental services and treatment records ensuring consistent pricing across all branches. The system features a Payment/Billing Module that allows receptionists to generate accurate bills based on treatment details and enables patients to view and make payments through supported methods. Additionally, the system provides HR and admin tools to manage employee records including salary and banking information and generate reports to assist in better management and decision making.

CHAPTER 2

METHODOLOGY

2.1 Chosen Conceptual Object-Oriented Programming

This project applies the Object-Oriented Programming (OOP) concept in the development of the Centralized Multi-Branch Clinic Management System for Putra Dental Clinic. Object-Oriented Programming is chosen because it provides a structured and systematic way to design and develop software systems. By organizing the system into classes and objects, the development process becomes easier to manage and understand, especially for a system that involves multiple users, clinic branches, and large amounts of data.

OOP allows data and functions that are related to be grouped together in one unit. For example, patient details, appointment information, and billing records are managed within their own classes. This improves system organization and helps developers clearly identify the responsibility of each class. When updates or improvements are required, changes can be made to specific classes without affecting the entire system. This reduces errors and makes the system easier to maintain in the long term.

The system uses encapsulation to protect sensitive information such as patient medical records, treatment history, and payment details. Encapsulation prevents direct access to data fields and ensures that data can only be modified through controlled methods. This improves system security and reduces the risk of incorrect data entry. Inheritance is applied by creating a general User class that is inherited by specific user roles such as Admin, Dentist, and Receptionist. This approach reduces code repetition and improves development efficiency.

Polymorphism allows the same function to perform different actions based on the role of the user, ensuring that each user only has access to the functions relevant to their responsibilities.

Overall, the use of Object-Oriented Programming improves system flexibility, reusability, security, and scalability. It also supports future system expansion, such as adding new clinic branches or new system features, without major changes to the existing system structure.

2.2 Project Planning

The project begins by working closely with Putra Dental Clinic to understand their current challenges and needs. During this stage, we learn about problems such as scattered patient records, manual appointment booking, slow billing, and poor coordination between their branches in Parit Raja, Pagoh, and Kluang. Once we understand what the clinic requires, we decide what features the new system should have, such as online appointment scheduling, central patient records, digital billing, and secure login for different staff roles like dentists, receptionists, and administrators.

After setting the requirements, we move to the design phase. Here, we create diagrams such as activity diagrams, sequence diagrams, class diagrams and use case diagrams to plan how the system will work. These diagrams help us visualize the processes like how a patient books an appointment or how a receptionist issues a bill before any coding begins. This step ensures everyone on the team understands the system's structure and workflow clearly.

Once the design is approved, we start building the system. We develop the program using Java, applying object-oriented concepts to create modules for each function. We also test each part thoroughly to make sure everything works as expected. Finally, we prepare the system for demonstration and compile all our work into a complete project report, documenting every step from planning to final testing.

NAME	ROLE	RESPONSIBILITY
MUHAMMAD FARHAN BIN MAD ZIN	Project Leader / Programmer	Oversees project planning and progress, assigns tasks to team members, ensures objectives are met and acts as the main contact with the lecturer and stakeholder. Codes the main system functions, applies OOP concepts, and connects logic with the interface.
MUHAMMAD MUALIMIN BIN MUIE	Programmer	Develops the user interface for the dentists and HR staff, creating the screens and menus for their daily tasks.
AZRAEI AZKA BIN MOHD PAIZ	Programmer	Develops the user interface for the receptionists, creating the screens for booking appointments and managing patient check-ins.
DURROTUN NASIHAH BINTI MOHD FAIZAL	System Analyst	Talks to clinic staff to understand their needs and designs the system's workflow plans and diagrams.
FATIN DAYANA BINTI AMZAR KID	System Analyst	Designs the system's data structure and the visual wireframes for all the user interfaces to ensure a unified platform.

Table 1 : Roles of Each Group member

2.3 Gantt Chart

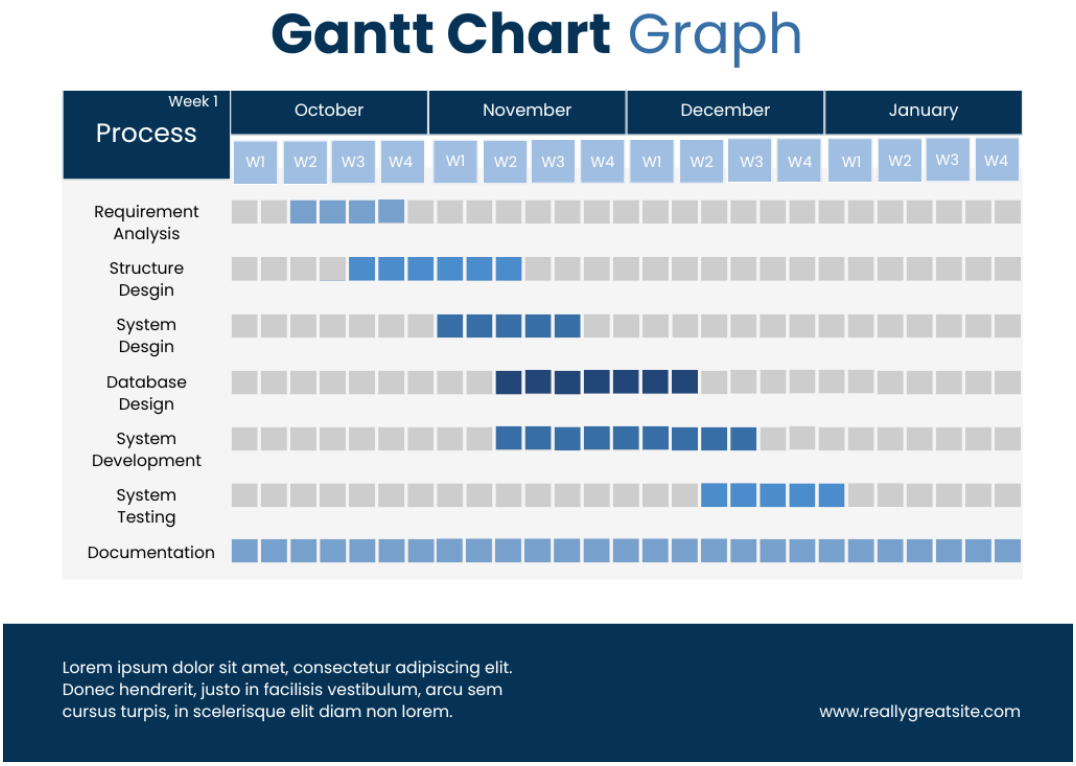


Figure 1 : Gantt Chart

2.4 Agile Methodology

We use Agile methodology for this project because it is flexible and helps us adapt to changes easily. Agile focuses on teamwork, regular updates, and continuous improvement. The project is divided into six sprints. Each sprint has a clear goal and set of tasks. The diagram below shows how each sprint connects to the next in a cycle.



Figure 2 : Agile Methodology Diagram

Sprint 1 : Planning

In this sprint we decide what the system should do. We list all features like patient registration, appointment booking, and branch management. We set up Eclipse, GitHub, XAMPP, and Tomcat on our computers. By the end we have a project plan and a working development environment.

Sprint 2 : Design

We design how the system will work. We draw diagrams to show the flow for patients, doctors, and admins. We design the database tables for patients, appointments, and branches using phpMyAdmin. We also sketch the website layout using HTML and CSS. By the end we have clear designs and a database structure.

Sprint 3 : Development

We start building the system. We code the backend in Java for features like adding patients and managing appointments. We create web pages using HTML, CSS, and JavaScript. We connect the website to the MySQL database. All code is saved in GitHub. By the end we have a working system with core features.

Sprint 4 : Testing

We test everything thoroughly. We check if patient registration works, if appointments can be booked, and if different branches can share data. We fix any bugs we find. We also test security to protect patient data. By the end we have a stable and reliable system.

Sprint 5 : Deployment

We install the system on a real server using Apache Tomcat. We set up the live database and make sure the system runs smoothly. We train staff on how to use it. By the end the system is ready for use in a clinic environment.

Sprint 6 : Review

We look back at the project. We check if we met our goals. We collect feedback from users and note down what can be improved. We prepare final reports and user guides. All code and documents are saved in GitHub. By the end we have a complete system and lessons for future projects.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Analysis

In the Analysis section, the system requirements and design considerations for the dental clinic management system are presented in detail. This part includes the Software Requirements Specification (SRS), which outlines the functional and non-functional requirements, and the Software Requirements Document (SRD), which translates these requirements into technical specifications. It also covers various UML diagrams that model the system's behavior and structure, including use case diagrams to represent user interactions, sequence diagrams to depict the flow of events between objects, activity diagrams to show process logic, and class diagrams to illustrate the entities, their attributes, and relationships. Together, these artifacts provide a comprehensive understanding of the system's functionality, workflows, and data structure, serving as the foundation for the subsequent design and implementation phases.

3.1.1 Requirement Specifications for Project Development

Module To Be Implemented	Description
Login Module	Admin, patient, dentist, receptionist and HR can login using username and password.
Appointment Scheduling Module	Receptionists schedule patient appointments with availability checks and booking confirmation.
Payment/Billing Module	Receptionists manage patient billing and payments, while patients can view their billing history.
Patient Record Management Module	Allows dentists and receptionists to view, register, search update patient records including personal details, contact information, medical notes and patient status.
Treatment Management Module	Allows admin create, view, update and delete treatment records and ensuring the system maintain an up-to-date list of services available for patients.
Report Generation Module	Admin generates and download revenue reports, showing invoice and payment details for all patients.
User Management Module	Admin manage user accounts including creation, updates and activation or deactivation of users.
Employee Management Module	HR manages employee records including personal, salary and banking information

Table 2 : Requirement Specifications Module

Functional Requirements:

- The system shall allow admin, patient, dentist, receptionist and HR to log in securely using a username and password.
- The system shall allow receptionist to schedule appointments and prevent double booking.

- The system shall allow receptionist to generate bills and record patient payments.
- The system shall allow patients to view their billing history.
- The system shall allow dentists and receptionists to view and update patient records.
- The system shall allow admins to manage treatment records.
- The system shall allow admins to generate revenue reports.
- The system shall allow admins to manage user accounts and access permissions.
- The system shall allow HR staff to manage employee records.

Non-Functional Requirements

- The system should restrict access based on user roles.
- The system should ensure data security and confidentiality.
- The system should provide acceptable response time for user requests.
- The system should support multiple concurrent users.
- The system should maintain data integrity during transactions.
- The system should be accessible via modern web browsers.

3.1.2 Use Case Diagram

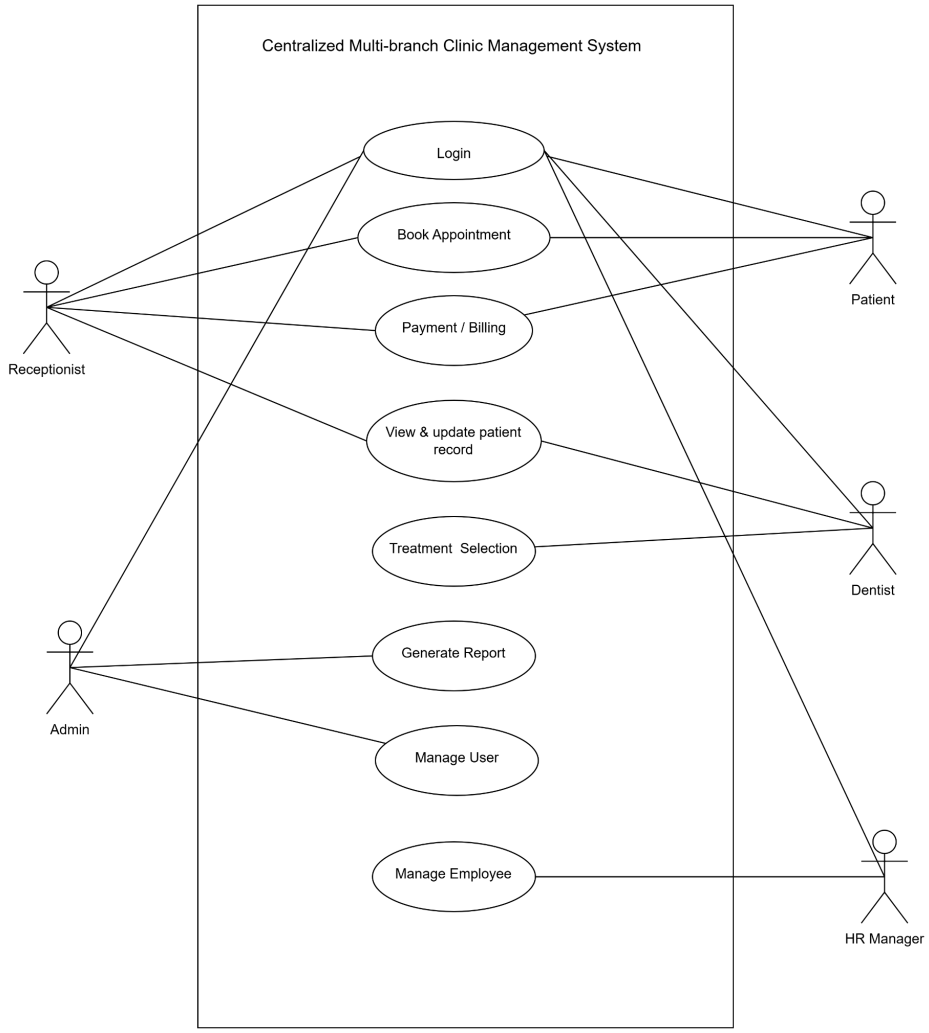


Figure 3 : Use Case Diagram of Centralized Multi-Branch Clinic Management System

This diagram represents a Centralized Multi-branch Clinic Management System. It shows how different users Receptionist, Patient, Dentist, Admin and HR Manager interact with the system. All users can log in, while core functions include booking appointments, payment/billing, viewing and updating patient records and treatment selection. Administrative roles handle report generation, user management, and employee management, ensuring smooth clinic operations across multiple branches.

3.1.3 Activity Diagram

1. Login Activity Diagram

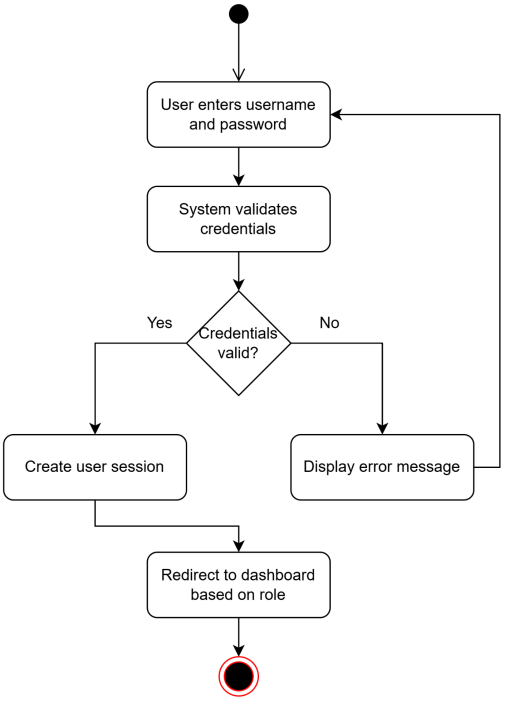


Figure 4 : Login Activity Diagram

The diagram illustrates the sequence of activities involved in a user login process. It begins with the user entering a username and password, after which the system validates the provided credentials. A decision node checks whether the credentials are valid, if they are, the system creates a user session and then redirects the user to the appropriate dashboard based on their role. If they are not, the system displays an error message and loops back to allow the user to re-enter their credentials, eventually leading to the end of the login activity.

2. Appointment Scheduling Activity Diagram

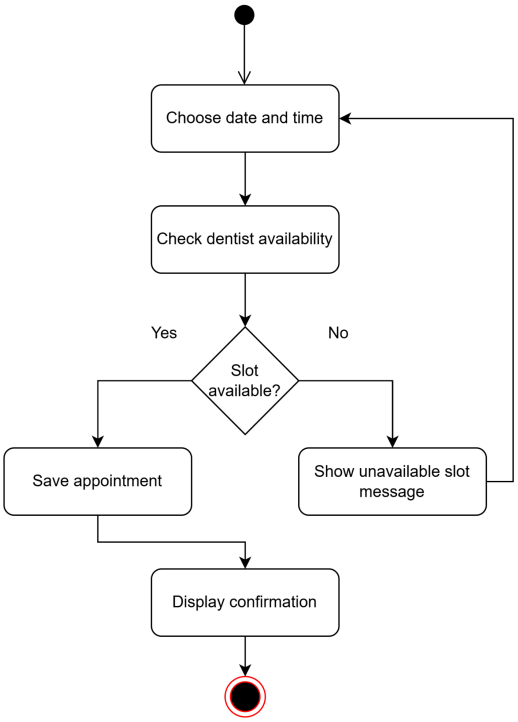


Figure 5 : Appointment Scheduling Activity Diagram

The diagram illustrates the process of scheduling a dentist appointment. It starts with the user choosing a date and time, followed by the system checking the dentist's availability for that slot. A decision node determines if the slot is available. If yes, the appointment is saved and a confirmation is displayed, leading to the end. If no, a message shows the slot is unavailable and loops back to allow choosing another date and time.

3. Payment/Billing Activity Diagram

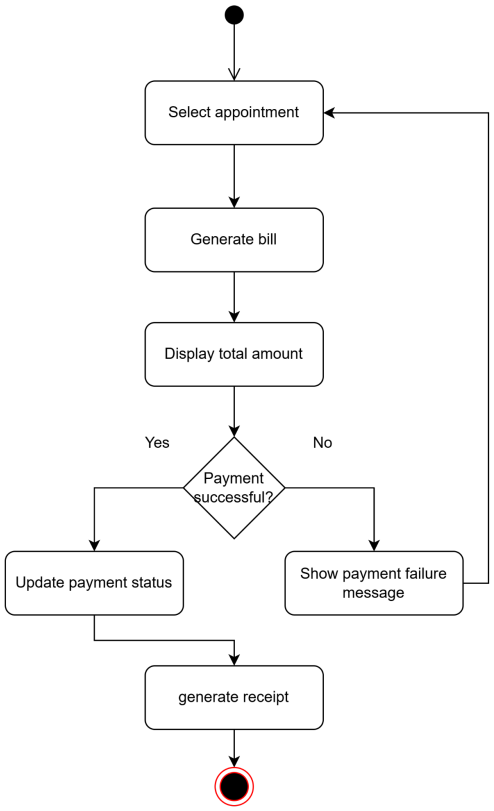


Figure 6 : Payment/Billing Activity Diagram

The diagram illustrates the payment and billing process for an appointment. It begins with selecting an appointment, generating a bill and displaying the total amount. A decision node checks if payment is successful. If yes, the payment status is updated and a receipt is generated, leading to the end. If no, a payment failure message is shown, allowing for a retry.

4. Patient Record Management Activity Diagram

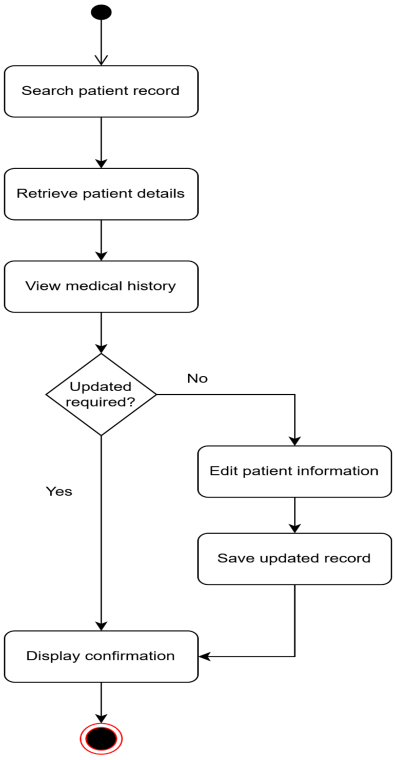


Figure 7 : Patient Record Management Activity Diagram

The diagram illustrates the workflow of a patient record management process within the system. The process begins when a user searches for a patient record, followed by retrieving the patient's details and viewing the medical history. After reviewing the information, a decision is made on whether an update is required. If no update is needed, the system directly displays a confirmation and ends the process. If an update is required, the user edits the patient information and saves the updated record. Once the changes are saved, the system displays a confirmation and the process is completed.

5. Treatment Management Activity Diagram

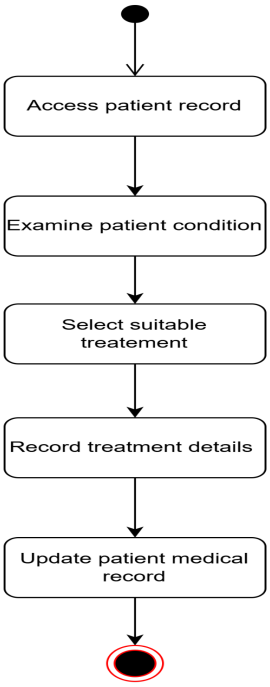


Figure 8 : Treatment Management Activity Diagram

The diagram illustrates the treatment management process within the system. The process starts when a dentist accesses the patient record and examines the patient's condition. Based on the examination, a suitable treatment is selected. The treatment details are then recorded, and the patient's medical record is updated accordingly. Finally, the process ends after the updated information has been successfully saved.

6. Report Generation Activity Diagram

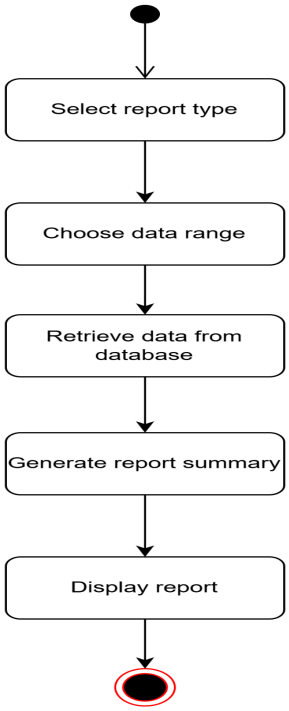


Figure 9 : Report Generation Activity Diagram

The diagram illustrates the report generation process within the system. The process begins when the user selects the type of report to be generated, followed by choosing the required data range. The system then retrieves the relevant data from the database and generates a report summary based on the selected criteria. Finally, the completed report is displayed to the user and the process comes to an end.

7. User Management Activity Diagram

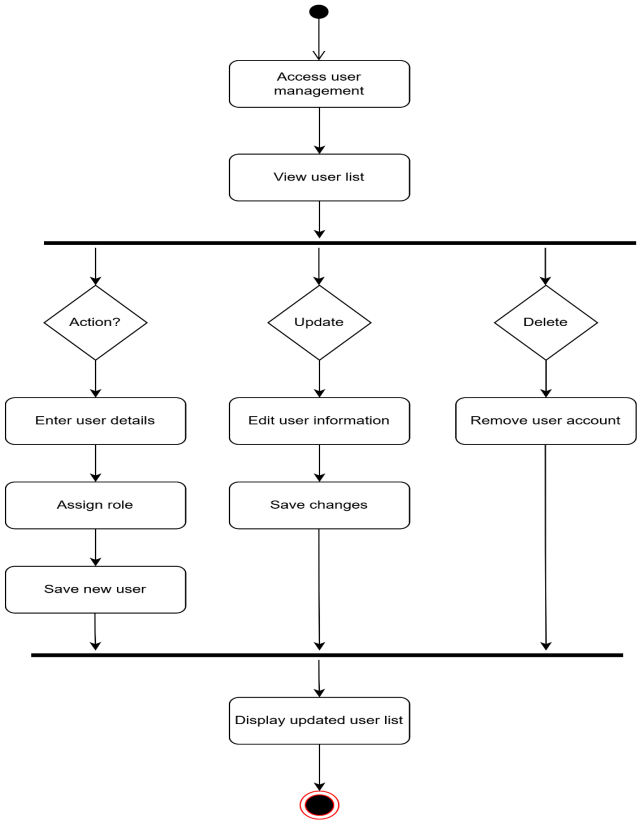


Figure 10 : User Management Activity Diagram

The diagram illustrates the user management activity flow within a system. The process begin when an administrator accesses the user management module and views the user list. From there, the process branches based on the selected action which is adding, updating or deleting a user. For adding a user, the administrator enters user details, assigns a role and saves the new user. For updating, the administrator edits existing user information and saves the changes. For deleting, the administrator removes the selected user account. All actions then converge to display the updated user list, after which the process ends.

8. Employee Management Activity Diagram

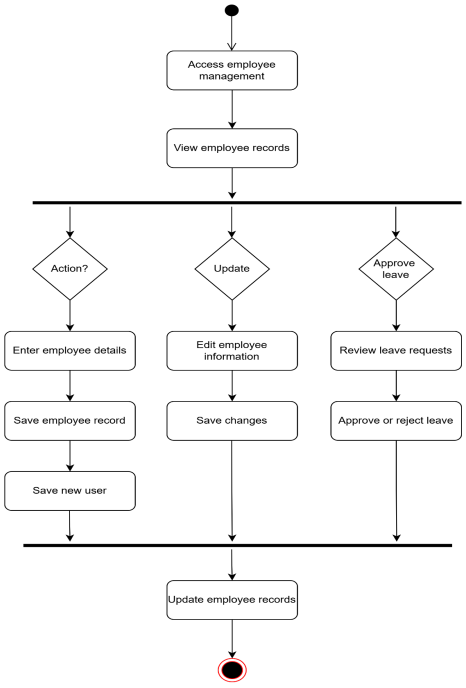


Figure 11 : Employee Management Activity Diagram

The diagram illustrates the employee management activity flow within a system. The process begin when an administrator accesses the employee management module and views employee records. The process then branches based on the selected action which is adding a new employee, updating employee information or approving leave requests. For adding an employee, the administrator enters employee details and saves the employee record. For updating, the administrator edits existing employee information and saves the changes. For leave management, the administrator reviews leave requests and either approves or rejects them. All paths converge to update the employee records, after which the process concludes.

3.1.4 Sequence Diagram

1. Login Sequence Diagram

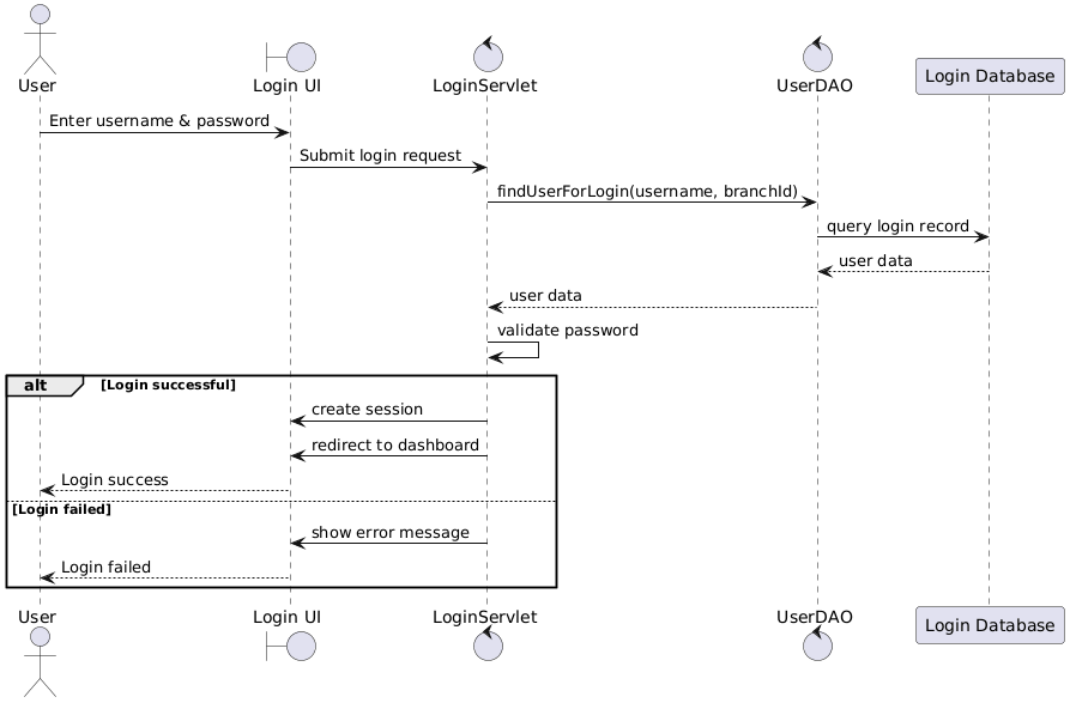


Figure 12 : Login Sequence Diagram

The login sequence starts when a user (Admin, Receptionist, Dentist, HR, or Patient) enters their credentials on the login page. The LoginController receives these credentials and passes them to the LoginService for validation. The service checks the user information against the Login Database. If the credentials match and the account is active, the service creates a session for the user and returns a success response. The controller then redirects the user to the appropriate dashboard based on their role. If the login fails due to incorrect credentials or inactive account, an error message is sent back to the user interface.

2. Appointment Scheduling Sequence Diagram

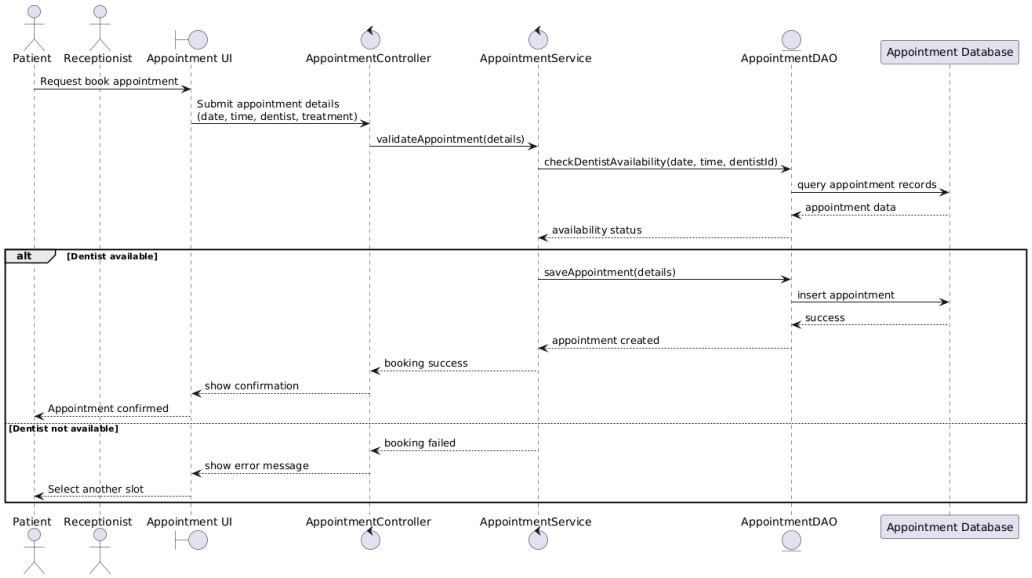


Figure 13 : Appointment Scheduling Sequence Diagram

Booking an appointment begins when a patient or receptionist submits the appointment details through the AppointmentController. The controller sends the data to the AppointmentService, which checks the availability of the chosen dentist and treatment via the AppointmentDAO. If the selected slot is available, the service saves the appointment to the Appointment Database and confirms success to the controller. The controller then shows a confirmation message to the patient or receptionist. If the slot is unavailable, an error message prompts the user to select another time.

3. Payment/Billing Sequence Diagram

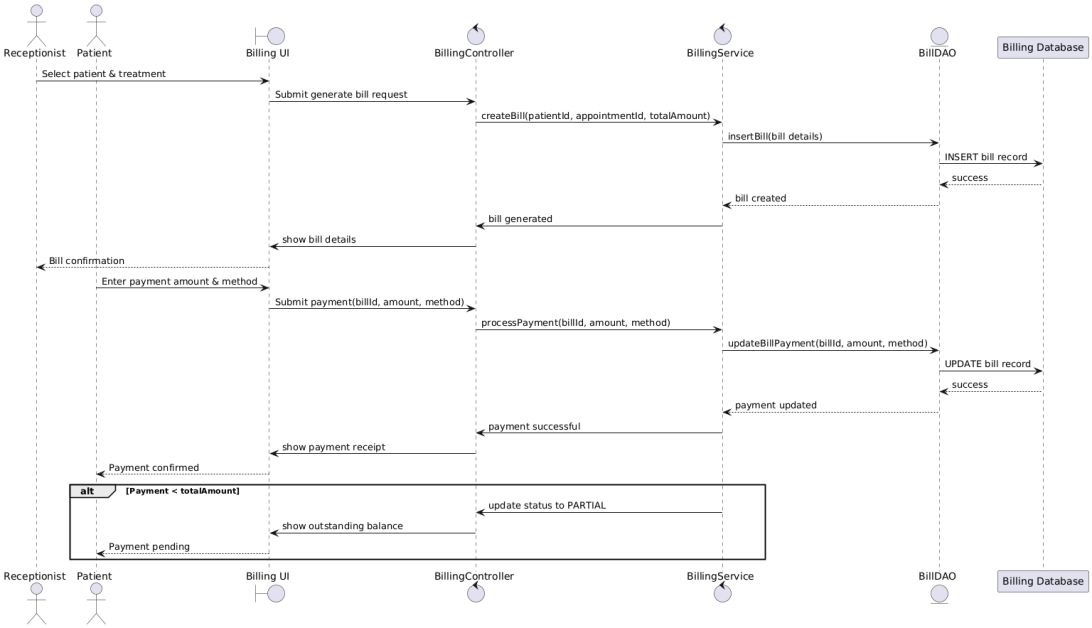


Figure 14 : Payment/Billing Sequence Diagram

The billing process is initiated when a patient completes a treatment. The BillingController receives the billing request and forwards it to the BillingService. The service calculates the total amount due and records the payment details using the BillingDAO. Payment can be made via cash, card, insurance, or online. After successful payment, the controller displays a confirmation receipt. If there is an issue, such as incomplete payment or invalid details, an error message is shown to the user.

4. Patient Record Management Sequence Diagram

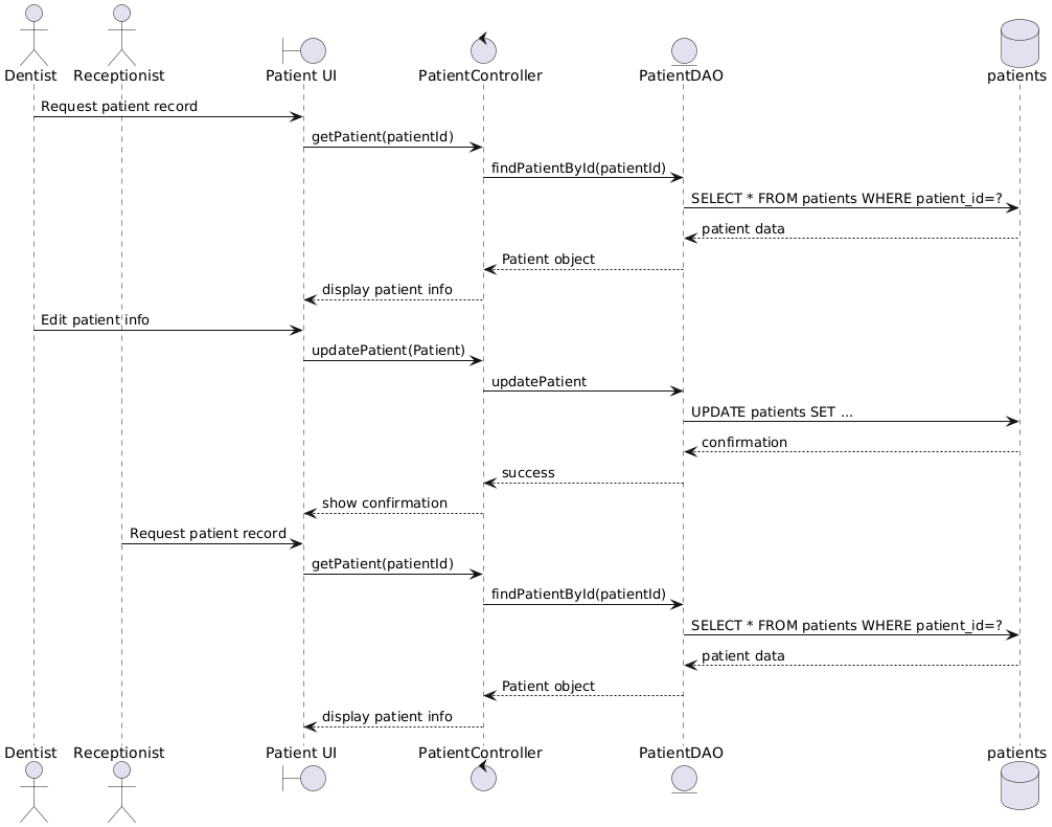


Figure 15: Patient Record Management Sequence Diagram

The Patient Record sequence diagram shows how the Dentist (and optionally the Receptionist) interacts with the system to view and update patient information. The actor initiates requests via the Patient UI, which sends them to the PatientController. For viewing records, the controller calls PatientDAO to retrieve patient data from the patients table, then returns it to the UI for display. For updates, modified patient data flows from the UI to the controller, then to the DAO for database update, and a confirmation is sent back to the UI. This process highlights separation of concerns, with the UI handling interaction, the controller managing logic, and the DAO isolating database operations, ensuring secure and structured access to patient information.

5. Treatment Management Sequence Diagram

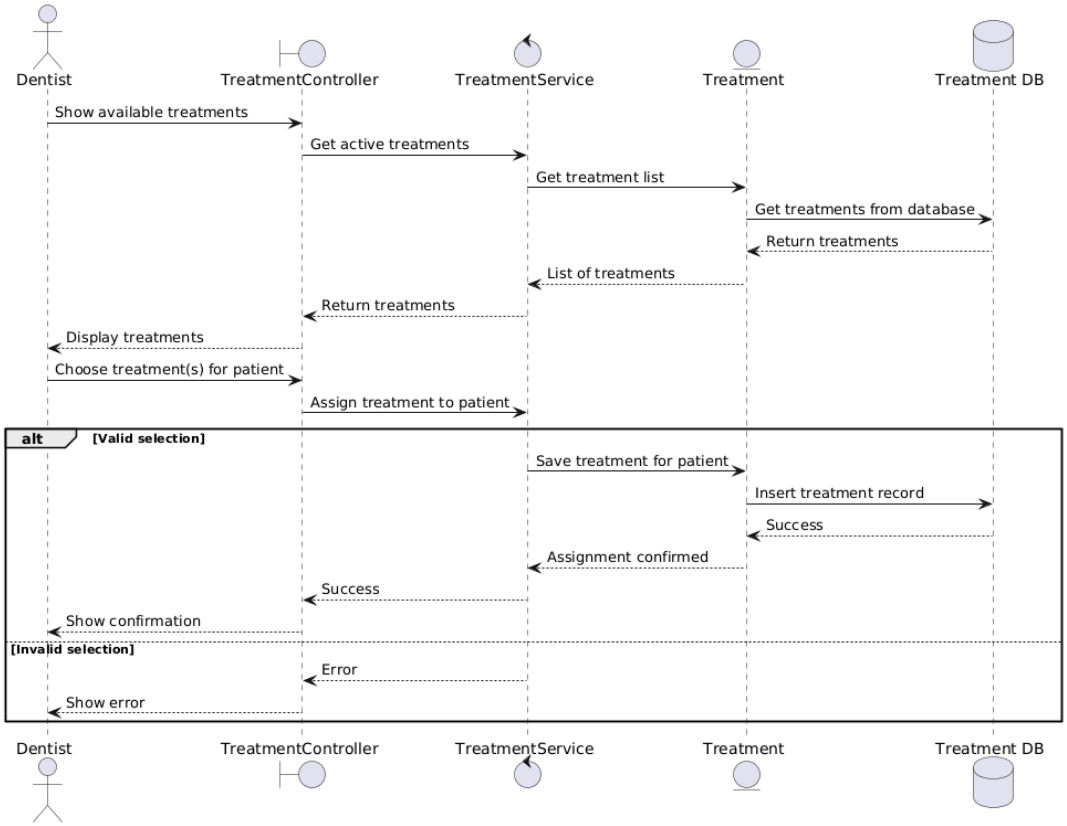


Figure 16: Treatment Management Sequence Diagram

During a consultation, the dentist selects appropriate treatments for the patient. The TreatmentController receives the selection and forwards it to the TreatmentService. The service records the selected treatments in the Treatment Database. After successful assignment, the controller shows a confirmation message. If the treatment selection is invalid, an error message is displayed to prompt the dentist to correct it.

6. Report Generation Sequence Diagram

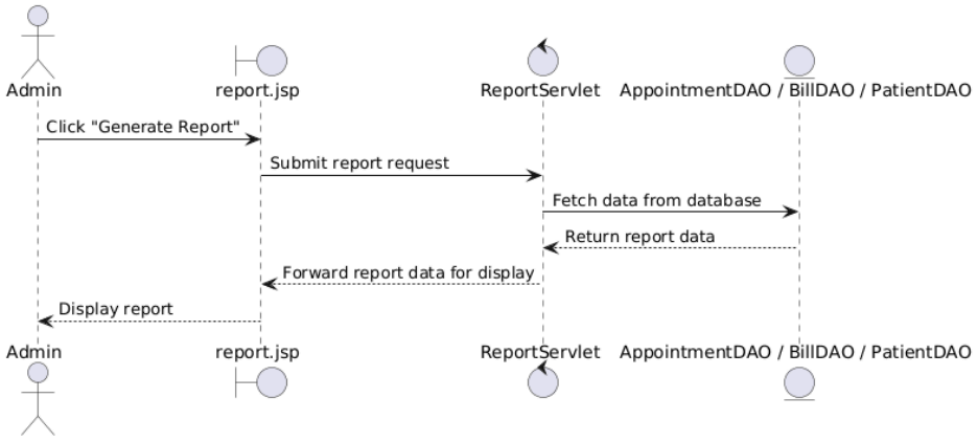


Figure 17 : Report Generation Sequence Diagram

The diagram illustrates how the system allows the admin to produce different reports, such as revenue, patient statistics or branch performance. The sequence begins with the admin interacting with the dashboard interface (JSP page) to request a specific report. The request is sent to the ReportServlet, which acts as the controller. The servlet invokes methods from the ReportDAO or similar data access class to retrieve data from the database. Once the data is fetched and processed, the servlet sends it back to the JSP for rendering in a readable format, such as tables or charts. This sequence ensures that the admin receives up-to-date and accurate information, helping with strategic decision-making.

7. User Management Sequence Diagram

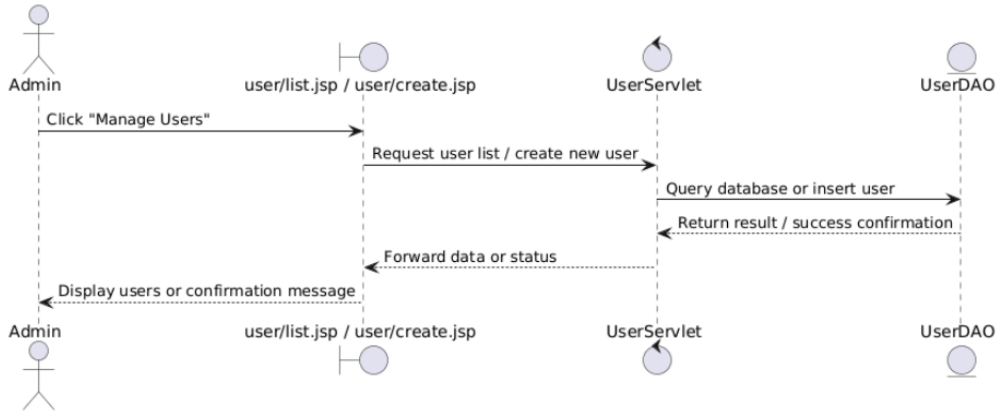


Figure 18 : User Management Sequence Diagram

The diagram illustrates how the system handles staff account management. The admin interacts with the interface (e.g., user/list.jsp or user/create.jsp) to either create a new user, edit existing user information, or deactivate a staff account. The request is sent to the UserServlet, which serves as the controller. Depending on the action, the servlet calls the UserDAO to perform database operations such as inserting a new user, updating user details or fetching user data. After the operation is complete, the servlet forwards the updated information back to the JSP page, providing feedback to the admin. This sequence ensures secure and centralized management of user accounts with role-based access control.

8. Employee Management Sequence Diagram

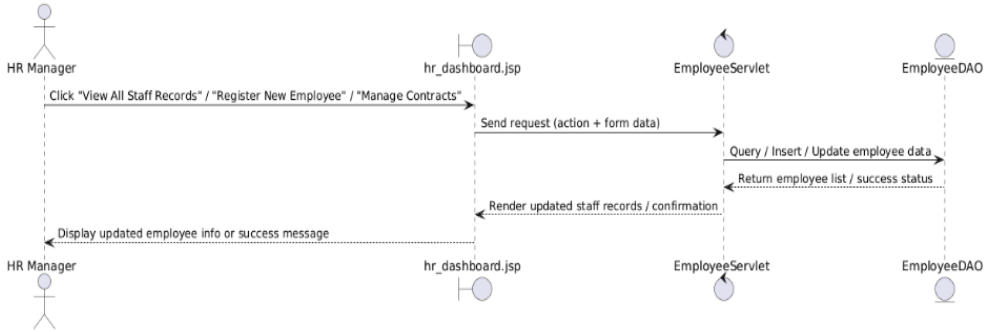


Figure 19 : Employee Management Sequence Diagram

The diagram illustrates how the HR officer manages employee records and information. The HR officer interacts with the interface (e.g., staff/records.jsp) to view, update, or add employee data. The request is captured by the HRServlet (or equivalent controller), which invokes EmployeeDAO methods to retrieve or update information in the database. Once the data is processed, the servlet returns the updated records to the JSP for display. This sequence ensures that employee information, including personal data, employment details, and leave records, is maintained accurately and efficiently, supporting HR operations and organizational management.

3.1.5 Class Diagram

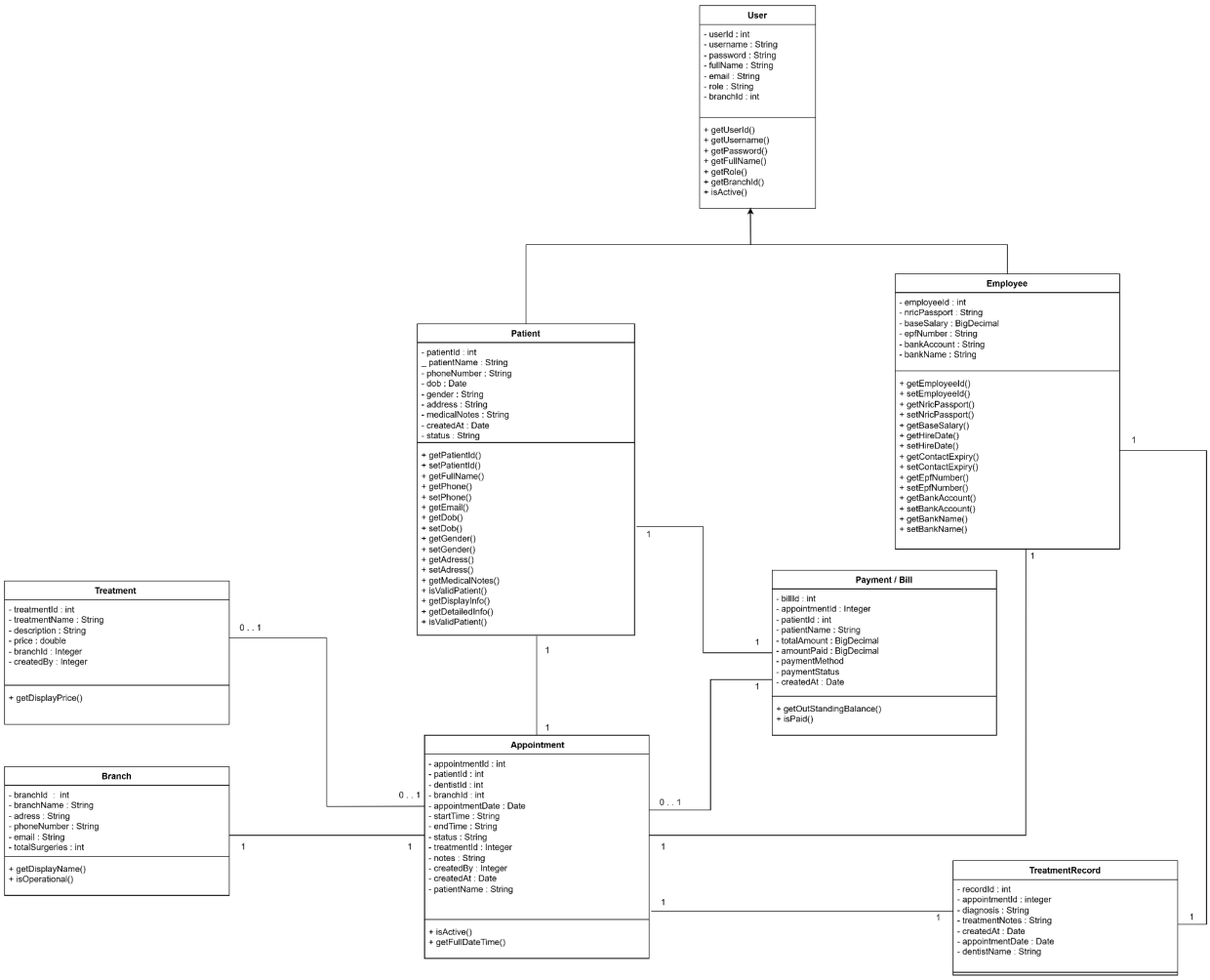


Figure 20: Class Diagram of Centralized Multi-Branch Clinic Management System

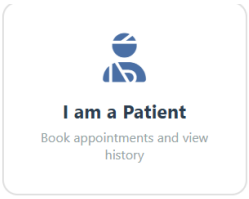
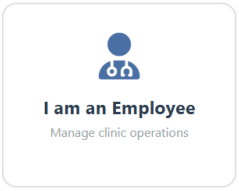
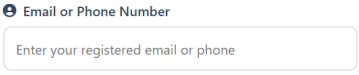
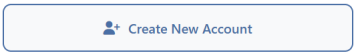
The class diagram models the key entities of the dental clinic system and their relationships. At the top of the hierarchy, the User class serves as a general template for all system users, providing common attributes such as userId, username, password, fullName, email, role, and branchId. Both Patient and Employee inherit from User, reflecting that patients and staff share these core properties while also having their own specific attributes. For example, patient includes personal details like phone, dob, gender, and medicalNotes, whereas an Employee includes employment-related information such as baseSalary, hireDate, and bankAccount. Other classes, including

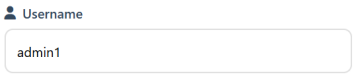
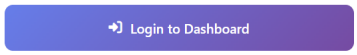
Appointment, Bill, Branch, Treatment, and TreatmentRecord, are associated with these user classes through various relationships, representing real-world interactions: appointments link patients, dentists, branches, and treatments; bills are connected to patients and appointments; and treatment records track the services provided by dentists.

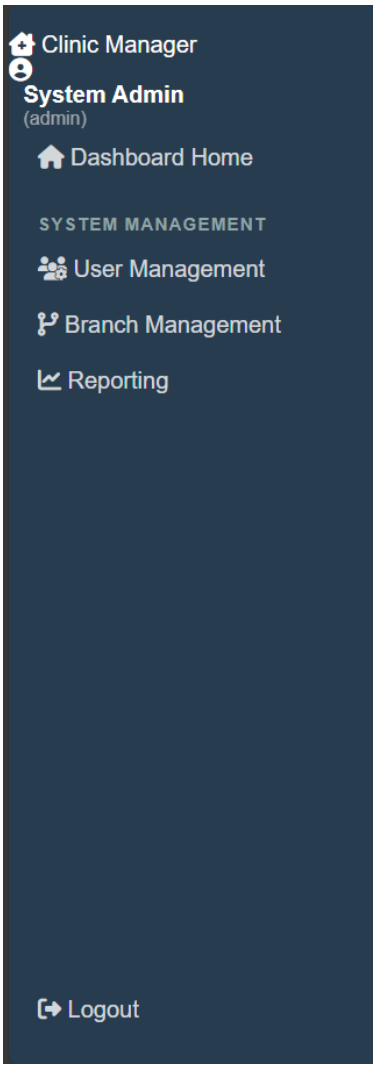
CHAPTER 4

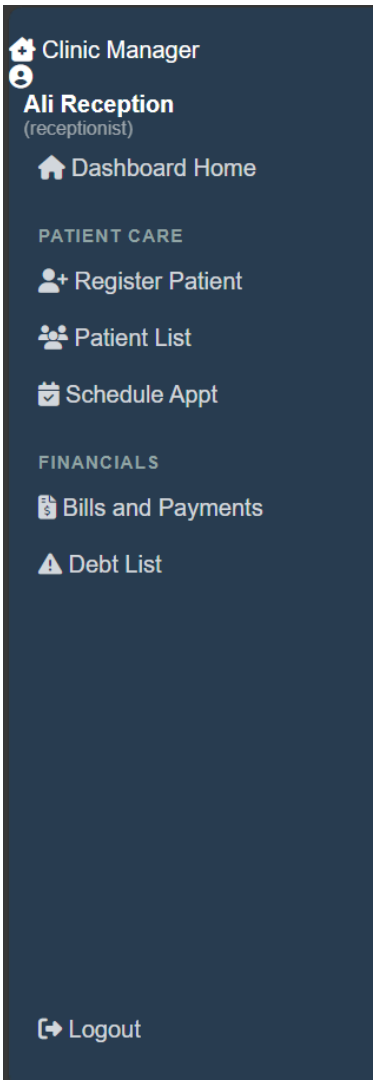
IMPLEMENTATION

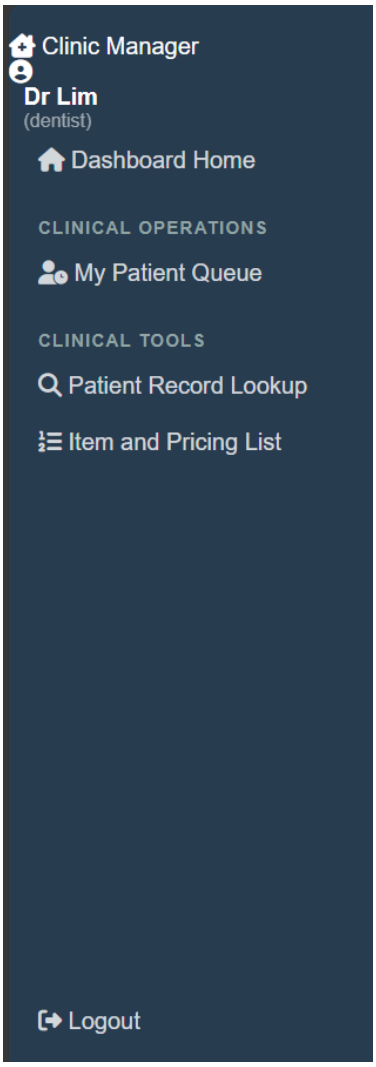
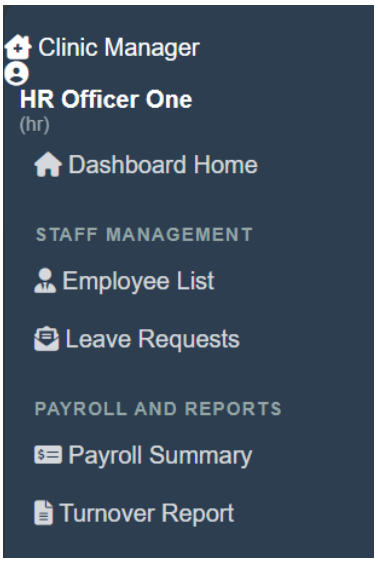
4.1 Input Design

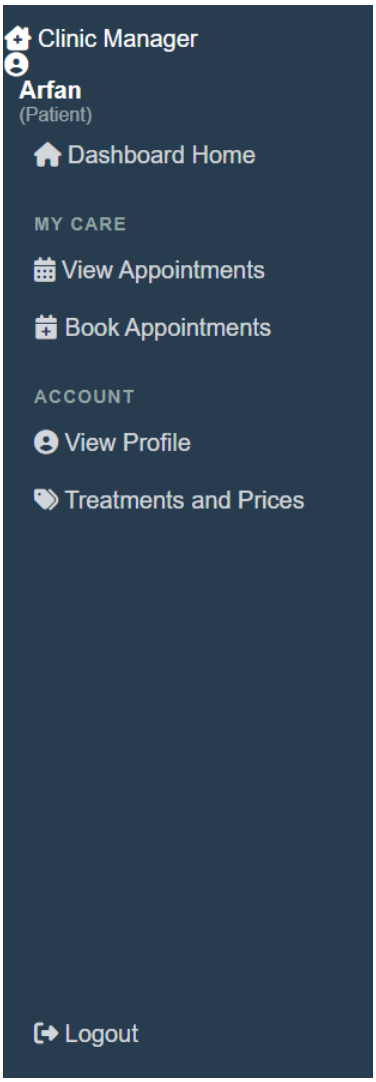
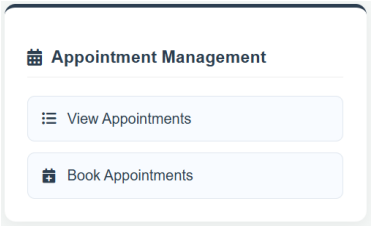
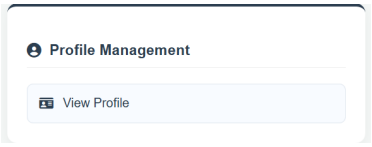
Name	Input	Type
I am a Patient button		Button
I am an Employee button		Button
Email or phone number box (patient)		Input box
Password Box (patient)		Password field
Log in button (patient)		Button
Register button (patient)		Button
Back button (patient)		Button

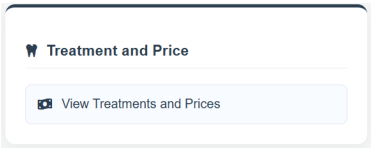
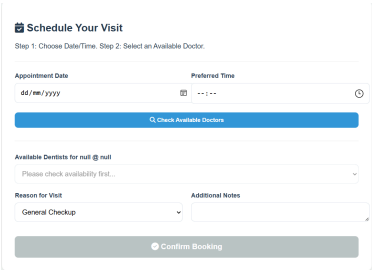
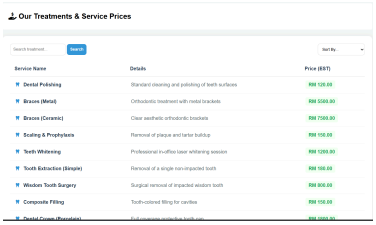
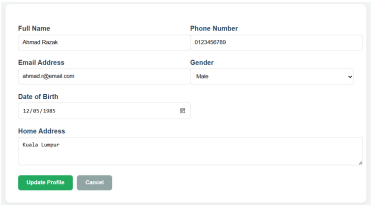
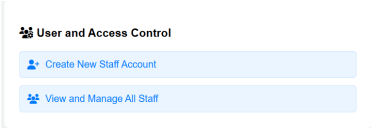
Username box (Employee)		Input box
Password Box (Employee)		Password field
Branch Box (Employee)		Dropdown
Login Button (Employee)		Button

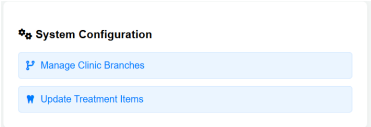
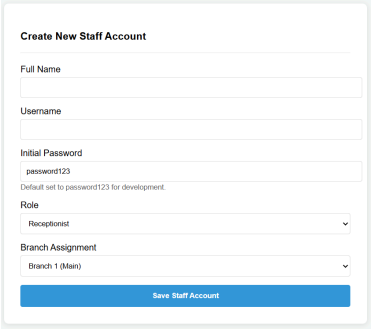
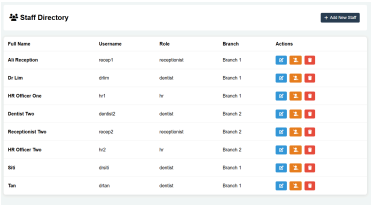
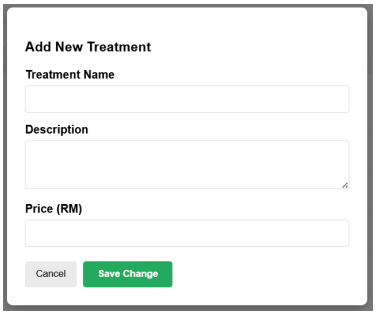
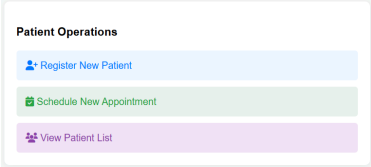
Dashboard Sidebar (Admin)		Sidebar Navigation

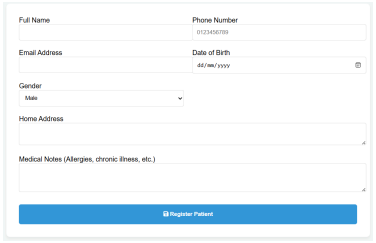
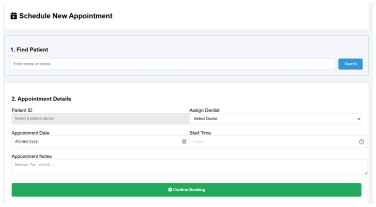
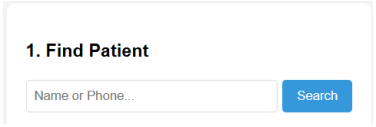
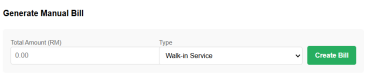
Dashboard Sidebar (Receptionist)		Sidebar Navigation

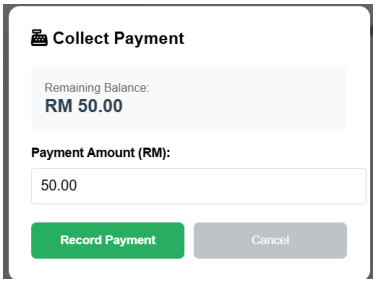
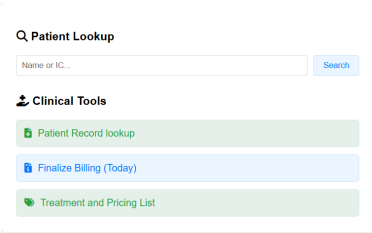
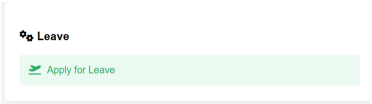
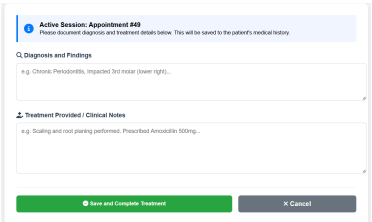
Dashboard Sidebar (Dentist)		Sidebar Navigation
Dashboard Sidebar (HR)		Sidebar Navigation

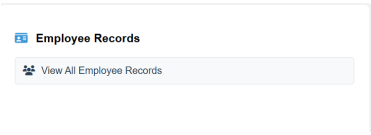
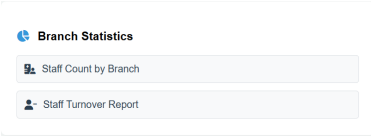
Dashboard Sidebar (Patient)		Sidebar Navigation
Appointment Management Container (Patient)		Button
Profile Management Container (Patient)		Button

Treatment and Price Container (Patient)		Button
Book Appointment Form (Patient)		Input box rand Button
Search and Sort view treatment (Patient)		Search Bar and drop down
Edit Profile (Patient)		Button
Edit Profile Form (Patient)		Input box, Drop Down and Button
Update Password Form (Patient)		Input Box and Button
User and Access Control Container (Admin)		Button

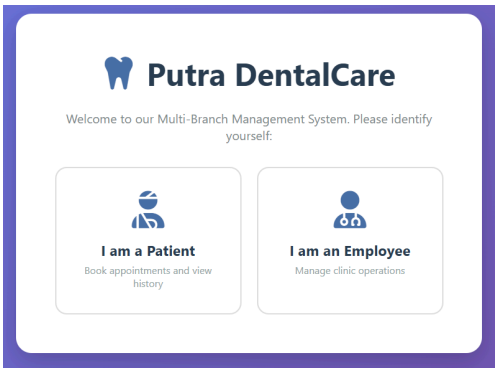
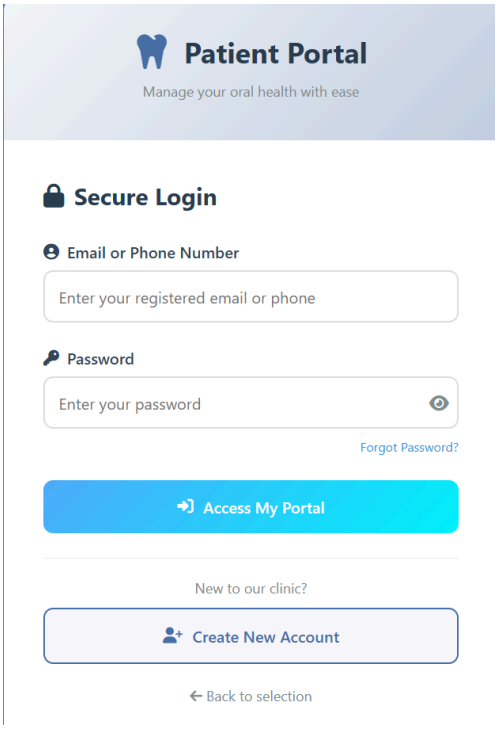
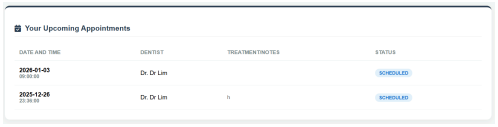
System Configuration Container (Admin)		Button
Create New Employee Account form (Admin)		Input box, Drop Down and Button
Manage Employee's Account (Admin)		Button
Treatment Management (Admin)		Button, Search bar, Drop down
Add New Treatment Form (Admin)		Input Box and Button
Treatment Edit Button (Admin)		Button
Patient Operations Container (Receptionist)		Button

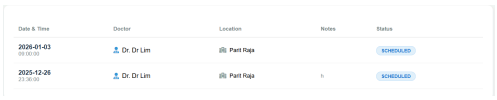
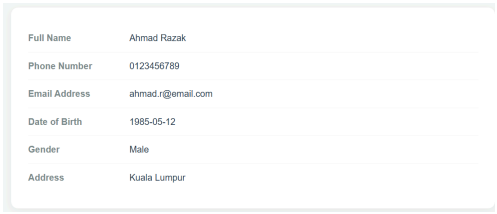
Finance and Leave Container (Receptionist)		Button
Cancel button (Receptionist)		Button
Register New Patient Form (Receptionist)		Input Box, Drop down Calendar and Button
Patient Appointment Booking Form (Receptionist)		Search Bar, Drop down and Button
Search Patient by name or phone (Receptionist)		Search Bar and Button
Find Patient for Invoice (Receptionist)		Search Bar and Button
Pay Button (Receptionist)		Button
Generate Manual Bill Form (Receptionist)		Input Box, Drop Down and Button

Collect Payment (Receptionist)		Button
Collect Payment Form (Receptionist)		Input box and Button
Patient Lookup And Clinical Tools Container (Dentist)		Search Bar and Button
Apply Leave Container (Dentist)		Button
View Patient History (Dentist)		Button
Treatment Search and Filter (Dentist)		Search Bar, Button and Drop Down
Treat Patient Button (Dentist)		Button
Treatment Form (Dentist)		Input Box and Button

Complete Registration Button (HR)		Button
Employee Records Container (HR)		Button
Edit and Deactivate Button (HR)		Button
Attendance and Leave Container (HR)		Button
Staff Count Container (HR)		Button

4.2 Output Design

Name	Output
Start Page	
Patient Login Form (Patient)	
Upcoming Appointment List (Patient)	

View Appointmnet (Patient)	
View Profile (Patient)	

CHAPTER 5

RESULT AND CONCLUSION

4.3 Result Outcome

The development of the Putra Dental Clinic Management System is expected to deliver significant functional and operational improvements for both patients and clinic staff across every branch. On the patient side, the system provides a more convenient and efficient way to manage their oral healthcare through online appointment booking. This reduces the dependency on physical records and eliminates repeated action for manual data entry at every visit. As a result, patients experience improved service accessibility, fewer missed appointments due to better scheduling clarity, and a smoother experience when visiting different clinic branches.

For the clinic staff and admin, the system minimizes manual paperwork by automating patient registration, medical record storage, and appointment management. Dentists can quickly access accurate and up-to-date patient histories so dentists can provide better and faster dental treatment without the risk of misleading information or unprepared treatment procedures. The system's reporting features summarizes from the patient credentials to the dental treatment billing and price. Inventory planning for medical supplies and medicine also becomes more efficient as the usage trends can be monitored through the centralized system. Overall, the system improves workflow efficiency, reduces human error when using manual record keeping and ensures smoother clinic operations across every branch.

4.4 Conclusion

The Putra Dental Clinic Management System successfully addresses the critical limitations of the clinic's current manual and fragmented processes by providing a centralized, user-friendly digital platform to manage oral healthcare services. By integrating modules for secure login, real-time appointment scheduling, patient record management, and automated billing, the system enhances communication between patients and the clinic staff while significantly reducing the administrative burdens on doctors, receptionists, and HR personnel. The platform provides accurate storage and real-time retrieval of critical dental records and treatment histories across the Parit Raja, Pagoh, and Kluang branches which contributes to improved service quality and reduces the likelihood of misleading information.

In addition, the system allows clinic administrators and HR teams to efficiently manage medicine standardization, staff records, patient records, and operations across multiple branches while supporting data-driven decisions through automated reports. With these integrated features, the Putra Dental Clinic Management System enhances operational efficiency, promotes consistent preventive care, improves patient satisfaction, and also supports a more structured and modern dental service environment. Overall, the project successfully meets its primary objectives and establishes a scalable foundation for future improvements as Putra Dental Clinic continues to expand its presence in Johor.